

CorGe +

Developed By- Michigan Department of Health & Human Services

Advanced Molecular Detection Southeast Region

This resource was made possible through funding provided under the Epidemiology and Laboratory Capacity for Prevention and Control of Emerging Infectious Diseases (ELC) Cooperative Agreement (CK24-0002), Project D: Advanced Molecular Detection to the Florida Department of Health. The conclusions, findings, and opinions expressed by authors do not necessarily reflect the official position of the U.S. Department of Health and Human Services, the Public Health Service, or the Centers for Disease Control and Prevention.

UPDATES

Complete 2026 Needs Assessment

- This year's assessment includes role-specific question paths, with certain sections tailored specifically for supervisors.
- To ensure complete representation, we are requesting one response from each of the following roles:
 - One supervisor
 - One epidemiologist
 - One bioinformatician
- Please submit all responses by **October 16, 2026**.

OVERVIEW

Purpose:

CorGe+, developed by the Michigan Department of Health & Human Services, provides high-throughput bacterial core-genome analysis, enabling clustering, linkage detection, cgMLST/SNP-based comparisons, phylogenetics, and metadata-integrated reporting for genomic surveillance and outbreak investigations.

Usage:

Used by public health laboratories and researchers to analyze bacterial genome assemblies, compare isolates against historical datasets, identify related clusters, generate Microreact visualizations, and prepare downstream PoODLE sheets for detailed SNP/pangenome analysis.

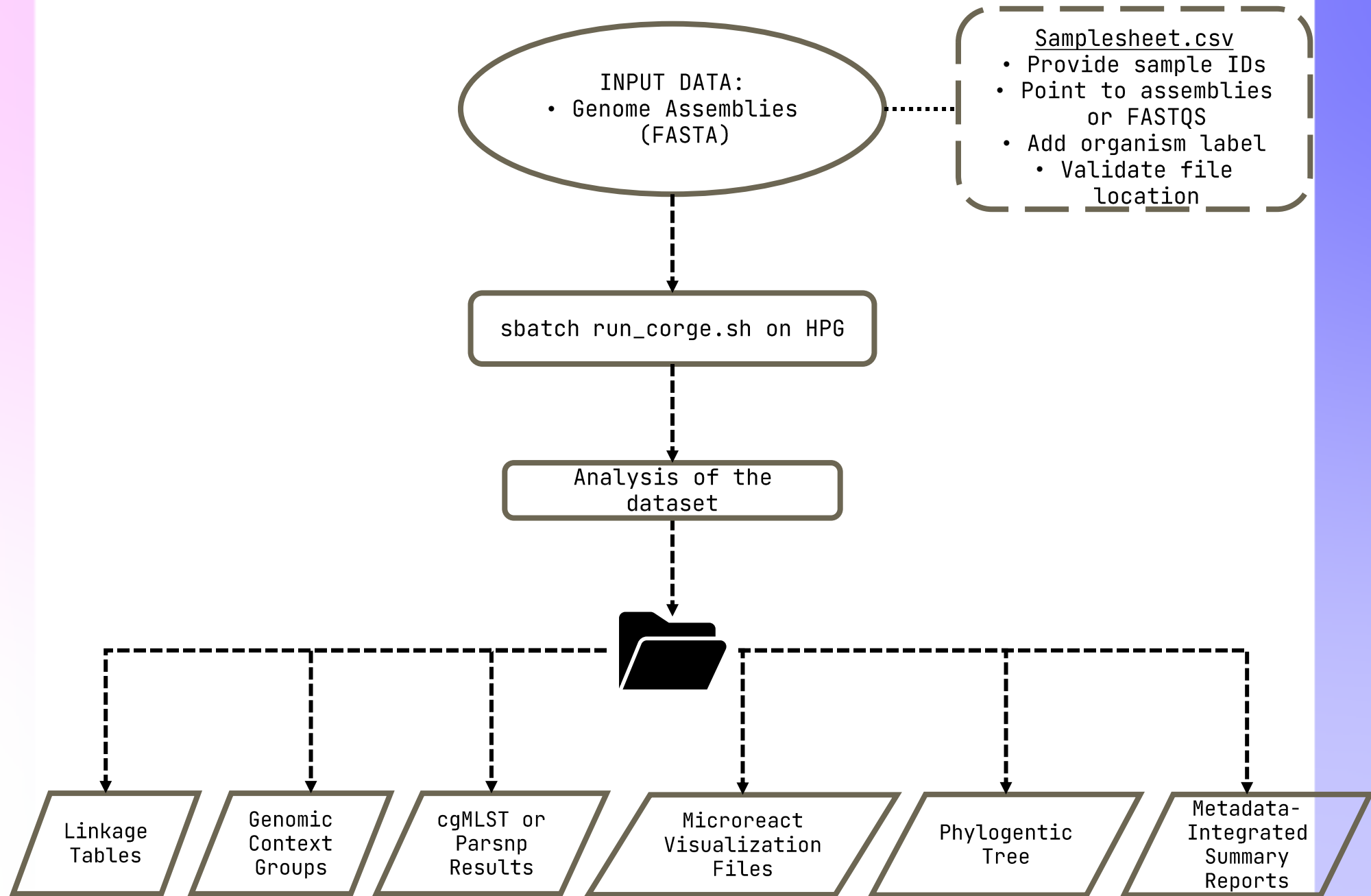
Dependencies:

Nextflow

Conda

Apptainer/Singularity

WORKFLOW



APPLICATION

OBJECTIVE

Use various genome assemblies (FASTA) and analyze the dataset using CorGe+ pipeline

APPLICATION CONT.

```
cd /blue/bphl-<state>/<user>/repos/bphl-molecular/  
git clone https://github.com/MDHHS-  
Bioinformatics/corge  
mkdir analysis/  
cd analysis/  
cp /blue/bphl-<state>/<user>/repos/bphl-molecular/  
corge/*
```

blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/corge0928/				
Name	Size	Changed	Rights	Owner
..		8/10/2026 11:16:18 AM	rw-r--r--	n.yengalareddy
assemblies		8/12/2026 1:27:45 PM	rw-r--r--	n.yengalareddy
assets		8/10/2026 11:14:55 AM	rw-r--r--	n.yengalareddy
bin		8/10/2026 11:14:55 AM	rw-r--r--	n.yengalareddy
cache		8/12/2026 1:48:56 PM	rw-r--r--	n.yengalareddy
conf		8/10/2026 11:14:55 AM	rw-r--r--	n.yengalareddy
docs		8/10/2026 11:14:55 AM	rw-r--r--	n.yengalareddy
fastq		8/12/2026 12:16:32 PM	rw-r--r--	n.yengalareddy
lib		8/10/2026 11:14:55 AM	rw-r--r--	n.yengalareddy
misc-data		8/12/2026 12:16:16 PM	rw-r--r--	n.yengalareddy
modules		8/10/2026 11:14:55 AM	rw-r--r--	n.yengalareddy
output-20260812164858		8/12/2026 1:31:52 PM	rw-r--r--	n.yengalareddy
subworkflows		8/10/2026 11:14:56 AM	rw-r--r--	n.yengalareddy
test		8/10/2026 11:14:56 AM	rw-r--r--	n.yengalareddy
training_files		8/10/2026 11:14:56 AM	rw-r--r--	n.yengalareddy
work		8/12/2026 1:48:55 PM	rw-r--r--	n.yengalareddy
workflows		8/10/2026 11:14:56 AM	rw-r--r--	n.yengalareddy
CHANGELOG.md	1 KB	8/10/2026 11:14:55 AM	rw-r--r--	n.yengalareddy
CITATION.cff	1 KB	8/10/2026 11:14:55 AM	rw-r--r--	n.yengalareddy
CITATIONS.md	6 KB	8/10/2026 11:14:55 AM	rw-r--r--	n.yengalareddy
CODE_OF_CONDUCT.md	3 KB	8/10/2026 11:14:55 AM	rw-r--r--	n.yengalareddy
CONTRIBUTING.md	5 KB	8/10/2026 11:14:55 AM	rw-r--r--	n.yengalareddy
CONTRIBUTORS.md	1 KB	8/10/2026 11:14:55 AM	rw-r--r--	n.yengalareddy
corge.39291391.err	1 KB	8/12/2026 1:31:36 PM	rw-r--r--	n.yengalareddy
corge.39291391.out	63 KB	8/12/2026 1:48:58 PM	rw-r--r--	n.yengalareddy
LICENSE	2 KB	8/10/2026 11:14:55 AM	rw-r--r--	n.yengalareddy
main.nf	3 KB	8/10/2026 11:14:55 AM	rw-r--r--	n.yengalareddy
modules.json	1 KB	8/10/2026 11:14:55 AM	rw-r--r--	n.yengalareddy
nextflow.config	11 KB	8/10/2026 11:14:56 AM	rw-r--r--	n.yengalareddy
nextflow_schema.json	25 KB	8/10/2026 11:14:56 AM	rw-r--r--	n.yengalareddy
pyproject.toml	1 KB	8/10/2026 11:14:56 AM	rw-r--r--	n.yengalareddy
README.md	16 KB	8/10/2026 11:14:55 AM	rw-r--r--	n.yengalareddy
run_corge.sh	1 KB	8/12/2026 1:30:54 PM	rw-r--r--	n.yengalareddy

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/corge0928/assemblies/				
Name	Size	Changed	Rights	Owner
..		8/12/2026 1:48:58 PM	rw-r--r--	n.yengalareddy
SRR40019073		8/12/2026 12:48:35 PM	rw-r--r--	n.yengalareddy
SRR40019074		8/12/2026 12:52:57 PM	rw-r--r--	n.yengalareddy
SRR40019075		8/12/2026 12:56:50 PM	rw-r--r--	n.yengalareddy
SRR40019078		8/12/2026 1:02:18 PM	rw-r--r--	n.yengalareddy
SRR40019080		8/12/2026 1:07:32 PM	rw-r--r--	n.yengalareddy
SRR40064142		8/12/2026 12:19:53 PM	rw-r--r--	n.yengalareddy
SRR40064153		8/12/2026 12:22:46 PM	rw-r--r--	n.yengalareddy
SRR40064187		8/12/2026 12:26:01 PM	rw-r--r--	n.yengalareddy
SRR40064547		8/12/2026 12:29:14 PM	rw-r--r--	n.yengalareddy
SRR40064574		8/12/2026 12:32:41 PM	rw-r--r--	n.yengalareddy
SRR40077207		8/12/2026 12:34:42 PM	rw-r--r--	n.yengalareddy
SRR40077208		8/12/2026 12:36:42 PM	rw-r--r--	n.yengalareddy
SRR40077209		8/12/2026 12:38:39 PM	rw-r--r--	n.yengalareddy
SRR40077210		8/12/2026 12:40:38 PM	rw-r--r--	n.yengalareddy
SRR40077212		8/12/2026 12:42:55 PM	rw-r--r--	n.yengalareddy
assemble.sh	2 KB	8/12/2026 12:06:34 PM	rw-r--r--	n.yengalareddy
assemblies.39286047.err	265 KB	8/12/2026 1:07:32 PM	rw-r--r--	n.yengalareddy
assemblies.39286047.out	7,361 KB	8/12/2026 1:07:31 PM	rw-r--r--	n.yengalareddy
samplesheet.csv	2 KB	8/12/2026 1:28:13 PM	rw-r--r--	n.yengalareddy
SRR40019073.fasta	4,842 KB	8/12/2026 12:48:35 PM	rw-r--r--	n.yengalareddy
SRR40019074.fasta	4,710 KB	8/12/2026 12:52:57 PM	rw-r--r--	n.yengalareddy
SRR40019075.fasta	4,790 KB	8/12/2026 12:56:50 PM	rw-r--r--	n.yengalareddy
SRR40019078.fasta	4,760 KB	8/12/2026 1:02:18 PM	rw-r--r--	n.yengalareddy
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SRR40064547.fasta	3,994 KB	8/12/2026 12:29:14 PM	rw-r--r--	n.yengalareddy
SRR40064574.fasta	4,222 KB	8/12/2026 12:32:41 PM	rw-r--r--	n.yengalareddy
SRR40077207.fasta	5,434 KB	8/12/2026 12:34:42 PM	rw-r--r--	n.yengalareddy
SRR40077208.fasta	5,386 KB	8/12/2026 12:36:42 PM	rw-r--r--	n.yengalareddy
SRR40077209.fasta	5,275 KB	8/12/2026 12:38:39 PM	rw-r--r--	n.yengalareddy
SRR40077210.fasta	5,500 KB	8/12/2026 12:40:38 PM	rw-r--r--	n.yengalareddy
SRR40077212.fasta	5,348 KB	8/12/2026 12:42:55 PM	rw-r--r--	n.yengalareddy

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SRR40077212		8/12/2026 12:42:55 PM	rw-r--r--	n.yengalareddy
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assemblies.39286047.out	7,361 KB	8/12/2026 1:07:31 PM	rw-r--r--	n.yengalareddy
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SRR40077210.fasta	5,500 KB	8/12/2026 12:40:38 PM	rw-r--r--	n.yengalareddy
SRR40077212.fasta	5,348 KB	8/12/2026 12:42:55 PM	rw-r--r--	n.yengalareddy

```
sample,assembly,species
SRR40019073,/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/corge0928/assemblies/SRR40019073.fasta,Shigella sonnei
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SRR40019075,/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/corge0928/assemblies/SRR40019075.fasta,Shigella sonnei
SRR40019078,/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/corge0928/assemblies/SRR40019078.fasta,Shigella sonnei
SRR40019080,/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/corge0928/assemblies/SRR40019080.fasta,Shigella sonnei
SRR40064142,/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/corge0928/assemblies/SRR40064142.fasta,Acinetobacter baumannii
SRR40064153,/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/corge0928/assemblies/SRR40064153.fasta,Acinetobacter baumannii
SRR40064187,/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/corge0928/assemblies/SRR40064187.fasta,Acinetobacter baumannii
SRR40064547,/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/corge0928/assemblies/SRR40064547.fasta,Acinetobacter baumannii
SRR40064574,/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/corge0928/assemblies/SRR40064574.fasta,Acinetobacter baumannii
SRR40077207,/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/corge0928/assemblies/SRR40077207.fasta,Escherichia coli
SRR40077208,/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/corge0928/assemblies/SRR40077208.fasta,Escherichia coli
SRR40077209,/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/corge0928/assemblies/SRR40077209.fasta,Escherichia coli
SRR40077210,/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/corge0928/assemblies/SRR40077210.fasta,Escherichia coli
SRR40077212,/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/corge0928/assemblies/SRR40077212.fasta,Escherichia coli
```

APPLICATION CONT.

blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/corge0928/				
Name	Size	Changed	Rights	Owner
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assemblies		8/12/2026 1:27:45 PM	rwxr-sr-x	n.yengalareddy
assets		8/10/2026 11:14:55 AM	rwxr-sr-x	n.yengalareddy
bin		8/10/2026 11:14:55 AM	rwxr-sr-x	n.yengalareddy
cache		8/12/2026 1:48:56 PM	rwxr-sr-x	n.yengalareddy
conf		8/10/2026 11:14:55 AM	rwxr-sr-x	n.yengalareddy
docs		8/10/2026 11:14:55 AM	rwxr-sr-x	n.yengalareddy
fastq		8/12/2026 12:16:32 PM	rwxr-sr-x	n.yengalareddy
lib		8/10/2026 11:14:55 AM	rwxr-sr-x	n.yengalareddy
misc-data		8/12/2026 12:16:16 PM	rwxr-sr-x	n.yengalareddy
modules		8/10/2026 11:14:55 AM	rwxr-sr-x	n.yengalareddy
output-20260812164858		8/12/2026 1:31:52 PM	rwxr-sr-x	n.yengalareddy
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workflows		8/10/2026 11:14:56 AM	rwxr-sr-x	n.yengalareddy
CHANGELOG.md	1 KB	8/10/2026 11:14:55 AM	rw-r--r--	n.yengalareddy
CITATION.cff	1 KB	8/10/2026 11:14:55 AM	rw-r--r--	n.yengalareddy
CITATIONS.md	6 KB	8/10/2026 11:14:55 AM	rw-r--r--	n.yengalareddy
CODE_OF_CONDUCT.md	3 KB	8/10/2026 11:14:55 AM	rw-r--r--	n.yengalareddy
CONTRIBUTING.md	5 KB	8/10/2026 11:14:55 AM	rw-r--r--	n.yengalareddy
CONTRIBUTORS.md	1 KB	8/10/2026 11:14:55 AM	rw-r--r--	n.yengalareddy
corge.39291391.err	1 KB	8/12/2026 1:31:36 PM	rw-r--r--	n.yengalareddy
corge.39291391.out	63 KB	8/12/2026 1:48:58 PM	rw-r--r--	n.yengalareddy
LICENSE	2 KB	8/10/2026 11:14:55 AM	rw-r--r--	n.yengalareddy
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nextflow.config	11 KB	8/10/2026 11:14:56 AM	rw-r--r--	n.yengalareddy
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run_corge.sh	1 KB	8/12/2026 1:30:54 PM	rw-r--r--	n.yengalareddy

```
#!/usr/bin/bash

#SBATCH --account=bphl-umbrella
#SBATCH --qos=bphl-umbrella
#SBATCH --job-name=corge_run
#SBATCH --ntasks=1
#SBATCH --cpus-per-task=40
#SBATCH --mem=200gb
#SBATCH --time=48:00:00
#SBATCH --output=corge.%j.out
#SBATCH --error=corge.%j.err
#SBATCH --mail-user=Nikhil.Yengala@flhealth.gov
#SBATCH --mail-type=FAIL,END

module load java
module load conda
module load apptainer
module load nextflow/25.10.4

# Apptainer container cache directory
export NXF_APPTAINER_CACHEDIR=/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/corge0928/cache/

# Move into project directory
cd /blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/corge0928/

# Run CorGe+
nextflow run MDHHS-Bioinformatics/corge \
  --input /blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/corge0928/assemblies/samplesheet.csv \
  --outdir output \
  --profile apptainer

# Timestamp output folder
dt=$(date "+%Y%m%d%H%M%S")
mv output "output-${dt}"
```

APPLICATION CONT.

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/corge0928/output-20260812164858/

Name	Size	Changed	Rights	Owner
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Acinetobacter_baumannii		8/12/2026 1:48:55 PM	rwxr-sr-x	n.yengalareddy
Escherichia_coli		8/12/2026 1:48:54 PM	rwxr-sr-x	n.yengalareddy
pipeline_info		8/12/2026 1:48:58 PM	rwxr-sr-x	n.yengalareddy
Shigella_sonnei		8/12/2026 1:48:53 PM	rwxr-sr-x	n.yengalareddy

blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/corge0928/

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lib		8/10/2026 11:14:55 AM	rwxr-sr-x	n.yengalareddy
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workflows		8/10/2026 11:14:56 AM	rwxr-sr-x	n.yengalareddy
CHANGELOG.md	1 KB	8/10/2026 11:14:55 AM	rw-r--r--	n.yengalareddy
CITATION.cff	1 KB	8/10/2026 11:14:55 AM	rw-r--r--	n.yengalareddy
CITATIONS.md	6 KB	8/10/2026 11:14:55 AM	rw-r--r--	n.yengalareddy
CODE_OF_CONDUCT.md	3 KB	8/10/2026 11:14:55 AM	rw-r--r--	n.yengalareddy
CONTRIBUTING.md	5 KB	8/10/2026 11:14:55 AM	rw-r--r--	n.yengalareddy
CONTRIBUTORS.md	1 KB	8/10/2026 11:14:55 AM	rw-r--r--	n.yengalareddy
corge.39291391.err	1 KB	8/12/2026 1:31:36 PM	rw-r--r--	n.yengalareddy
corge.39291391.out	63 KB	8/12/2026 1:48:58 PM	rw-r--r--	n.yengalareddy
LICENSE	2 KB	8/10/2026 11:14:55 AM	rw-r--r--	n.yengalareddy
main.nf	3 KB	8/10/2026 11:14:55 AM	rw-r--r--	n.yengalareddy
modules.json	1 KB	8/10/2026 11:14:55 AM	rw-r--r--	n.yengalareddy
nextflow.config	11 KB	8/10/2026 11:14:56 AM	rw-r--r--	n.yengalareddy
nextflow_schema.json	25 KB	8/10/2026 11:14:56 AM	rw-r--r--	n.yengalareddy
pyproject.toml	1 KB	8/10/2026 11:14:56 AM	rw-r--r--	n.yengalareddy
README.md	16 KB	8/10/2026 11:14:55 AM	rw-r--r--	n.yengalareddy
run_corge.sh	1 KB	8/12/2026 1:30:54 PM	rw-r--r--	n.yengalareddy

sbatch run_corge.sh

APPLICATION CONT.

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/corge0928/output-20260812164858/				
Name	Size	Changed	Rights	Owner
..		8/12/2026 1:48:58 PM	rwxr-sr-x	n.yengalareddy
Acinetobacter_baumannii		8/12/2026 1:48:55 PM	rwxr-sr-x	n.yengalareddy
Escherichia_coli		8/12/2026 1:48:54 PM	rwxr-sr-x	n.yengalareddy
pipeline_info		8/12/2026 1:48:58 PM	rwxr-sr-x	n.yengalareddy
Shigella_sonnei		8/12/2026 1:48:53 PM	rwxr-sr-x	n.yengalareddy

/.../bphl-molecular/analysis/corge0928/output-20260812164858/pipeline_info/		
Name	Size	Changed
..		8/12/2026 1:31:52 PM
execution_report_2026-08-12_16...	2,980 KB	8/12/2026 1:48:58 PM
execution_timeline_2026-08-12_...	260 KB	8/12/2026 1:48:58 PM
execution_trace_2026-08-12_16-...	8 KB	8/12/2026 1:48:57 PM
pipeline_dag_2026-08-12_16-31...	5 KB	8/12/2026 1:48:57 PM
samplesheet.valid.csv	3 KB	8/12/2026 1:31:52 PM
software_versions.yml	1 KB	8/12/2026 1:48:57 PM

Nextflow workflow report

[nasty_brazil]

Workflow execution completed successfully!

Run times
12-Aug-2026 16:31:44 - 12-Aug-2026 16:48:57 (duration: **17m 13s**)

47 succeeded

Nextflow command
nextflow run MDHHS-Bioinformatics/corge --input /blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/corge0928/assemblies/samplesheet.csv --outdir output -profile apptainer

CPU-Hours	0.4
Launch directory	/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/corge0928
Work directory	/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/corge0928/work
Project directory	/blue/bphl-florida/n.yengalareddy/cache/Nextflow/assets/MDHHS-Bioinformatics/corge
Script name	main.nf
Script ID	fd0314b84dbfa59d7b508dd6b70e573b
Workflow session	eb0fda12-5340-4fa8-8501-c049c7c7b6fe
Workflow repository	https://github.com/MDHHS-Bioinformatics/corge , revision: main (commit hash: a318180526576f7fe43936a566bf6278054ae65c)
Workflow profile	apptainer
Nextflow version	version 25.10.4, build 11173 (10-02-2026 15:17 UTC)

APPLICATION CONT.

/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/corge0928/output-20260812164858/

Name	Size	Changed	Rights	Owner
..		8/12/2026 1:48:58 PM	rwxr-sr-x	n.yengalareddy
Acinetobacter_baumannii		8/12/2026 1:48:55 PM	rwxr-sr-x	n.yengalareddy
Escherichia_coli		8/12/2026 1:48:54 PM	rwxr-sr-x	n.yengalareddy
pipeline_info		8/12/2026 1:48:58 PM	rwxr-sr-x	n.yengalareddy
Shigella_sonnei		8/12/2026 1:48:53 PM	rwxr-sr-x	n.yengalareddy

.../analysis/corge0928/output-20260812164858/Acinetobacter_baumannii/

Name	Size	Changed
..		8/12/2026 1:31:52 PM
assemblies		8/12/2026 1:31:52 PM
genomic_context_groups		8/12/2026 1:48:53 PM
linkages		8/12/2026 1:34:04 PM
mashtree		8/12/2026 1:37:56 PM
microreact		8/12/2026 1:48:51 PM
parsnp		8/12/2026 1:34:03 PM
poodle_samplesheets		8/12/2026 1:48:55 PM
ReporTree		8/12/2026 1:48:51 PM

APPLICATION CONT.

.../analysis/corge0928/output-20260812164858/Acinetobacter_baumannii/

Name	Size	Changed
..		8/12/2026 1:31:52 PM
assemblies		8/12/2026 1:31:52 PM
genomic_context_groups		8/12/2026 1:48:53 PM
linkages		8/12/2026 1:34:04 PM
mashtree		8/12/2026 1:37:56 PM
microreact		8/12/2026 1:48:51 PM
parsnp		8/12/2026 1:34:03 PM
poodle_samplesheets		8/12/2026 1:48:55 PM
ReportTree		8/12/2026 1:48:51 PM

.../corge0928/output-20260812164858/Acinetobacter_baumannii/assemblies/

Name	Size	Changed
..		8/12/2026 1:48:55 PM
SRR40064142.fna	3,894 KB	8/12/2026 1:31:52 PM
SRR40064153.fna	3,640 KB	8/12/2026 1:31:52 PM
SRR40064187.fna	4,094 KB	8/12/2026 1:31:52 PM
SRR40064547.fna	3,994 KB	8/12/2026 1:31:52 PM
SRR40064574.fna	4,222 KB	8/12/2026 1:31:52 PM

APPLICATION CONT.

Diagram showing a file hierarchy:

- Root: /.../output-20260812164858/Acinetobacter_baumannii/
- Subdirectory: genomic_context_groups/

Name	Size	Changed
..		8/12/2026 1:48:55 PM
Acinetobacter_baumannii-group...	1 KB	8/12/2026 1:48:53 PM
Acinetobacter_baumannii-group...	1 KB	8/12/2026 1:48:53 PM
Acinetobacter_baumannii-group...	1 KB	8/12/2026 1:48:53 PM
Acinetobacter_baumannii-group...	1 KB	8/12/2026 1:48:53 PM

Name	Size	Changed
..		8/12/2026 1:31:52 PM
assemblies		8/12/2026 1:31:52 PM
genomic_context_groups		8/12/2026 1:48:53 PM
linkages		8/12/2026 1:34:04 PM
mashtree		8/12/2026 1:37:56 PM
microreact		8/12/2026 1:48:51 PM
parsnp		8/12/2026 1:34:03 PM
poodle_samplesheets		8/12/2026 1:48:55 PM
ReporTree		8/12/2026 1:48:51 PM

sample,species,group_name,group_length,group_samples,report_date

SRR40064142	Acinetobacter_baumannii	HC0x1.0_singleton_5	1	SRR40064142	2026-08-12
SRR40064153	Acinetobacter_baumannii	HC0x1.0_singleton_1	1	SRR40064153	2026-08-12
SRR40064187	Acinetobacter_baumannii	HC0x1.0_singleton_4	1	SRR40064187	2026-08-12
SRR40064547	Acinetobacter_baumannii	HC0x1.0_singleton_2	1	SRR40064547	2026-08-12
SRR40064574	Acinetobacter_baumannii	HC0x1.0_singleton_3	1	SRR40064574	2026-08-12

APPLICATION CONT.

./../corge0928/output-20260812164858/Acinetobacter_baumannii/linkages/

Name	Size	Changed
..		8/12/2026 1:48:55 PM
Acinetobacter_baumannii_potent...	1 KB	8/12/2026 1:34:04 PM

../analysis/corge0928/output-20260812164858/Acinetobacter_baumannii/

Name	Size	Changed
..		8/12/2026 1:31:52 PM
assemblies		8/12/2026 1:31:52 PM
genomic_context_groups		8/12/2026 1:48:53 PM
linkages		8/12/2026 1:34:04 PM
mashtree		8/12/2026 1:37:56 PM
microreact		8/12/2026 1:48:51 PM
parsnp		8/12/2026 1:34:03 PM
poodle_samplesheets		8/12/2026 1:48:55 PM
ReporTree		8/12/2026 1:48:51 PM

sample,species,data_type,analysis_length,percentage_called,completeness_qc,min_dist,strong_linkages,intermediate_linkages,lineage_level

SRR40064142,Acinetobacter_baumannii,SNP,3909891,0.792,PASS,57,None,None,SRR40064547 (57)

SRR40064153,Acinetobacter_baumannii,SNP,3647158,0.849,PASS,51468,None,None,None

SRR40064187,Acinetobacter_baumannii,SNP,4116900,0.752,PASS,51512,None,None,None

SRR40064547,Acinetobacter_baumannii,SNP,4009310,0.772,PASS,57,None,None,SRR40064142 (57)

SRR40064574,Acinetobacter_baumannii,SNP,4232661,0.731,PASS,443,None,None,None

APPLICATION CONT.

Path: /.../corge0928/output-20260812164858/Acinetobacter_baumannii/mashtree/

Name	Size	Changed
..		8/12/2026 1:48:55 PM
Acinetobacter_baumannii.dnd	1 KB	8/12/2026 1:35:25 PM
Acinetobacter_baumannii.tsv	1 KB	8/12/2026 1:35:25 PM
Acinetobacter_baumannii_rooted...	1 KB	8/12/2026 1:37:56 PM

Path: .../analysis/corge0928/output-20260812164858/Acinetobacter_baumannii/

Name	Size	Changed
..		8/12/2026 1:31:52 PM
assemblies		8/12/2026 1:31:52 PM
genomic_context_groups		8/12/2026 1:48:53 PM
linkages		8/12/2026 1:34:04 PM
mashtree		8/12/2026 1:37:56 PM
microreact		8/12/2026 1:48:51 PM
parsnp		8/12/2026 1:34:03 PM
poodle_samplesheets		8/12/2026 1:48:55 PM
ReporTree		8/12/2026 1:48:51 PM

Distance matrix table:

	SRR40064574	SRR40064153	SRR40064187	SRR40064547	SRR40064142
SRR40064574	0	0.0230708	0.023519	0.00336684	0.00367873
SRR40064153	0.0230708	0	0.0224277	0.0214721	0.0207672
SRR40064187	0.023519	0.0224277	0	0.0223268	0.0218006
SRR40064547	0.00336684	0.0214721	0.0223268	0	0.00120098
SRR40064142	0.00367873	0.0207672	0.0218006	0.00120098	0

APPLICATION CONT.

.../analysis/corge0928/output-20260812164858/Acinetobacter_baumannii/

Name	Size	Changed
..		8/12/2026 1:31:52 PM
assemblies		8/12/2026 1:31:52 PM
genomic_context_groups		8/12/2026 1:48:53 PM
linkages		8/12/2026 1:34:04 PM
mashtree		8/12/2026 1:37:56 PM
microreact		8/12/2026 1:48:51 PM
parsnp		8/12/2026 1:34:03 PM
poodle_samplesheets		8/12/2026 1:48:55 PM
ReporTree		8/12/2026 1:48:51 PM

/.../corge0928/output-20260812164858/Acinetobacter_baumannii/microreact/

Name	Size	Changed
..		8/12/2026 1:48:55 PM
Acinetobacter_baumannii_corge....	205 KB	8/12/2026 1:48:51 PM

APPLICATION CONT.

File Explorer view showing the contents of the directory `/.../corge0928/output-20260812164858/Acinetobacter_baumannii/parsnp/`.

Name	Size	Changed
..		8/12/2026 1:48:55 PM
Acinetobacter_baumannii_parsn...	1,711 KB	8/12/2026 1:34:02 PM
Acinetobacter_baumannii_parsn...	15,297 KB	8/12/2026 1:34:03 PM
Acinetobacter_baumannii_parsn...	1 KB	8/12/2026 1:34:02 PM
Acinetobacter_baumannii_parsn...	467 KB	8/12/2026 1:34:02 PM
Acinetobacter_baumannii_parsn...	18,561 KB	8/12/2026 1:34:03 PM
Acinetobacter_baumannii_parsn...	1 KB	8/12/2026 1:34:02 PM
Acinetobacter_baumannii_parsn...	2 KB	8/12/2026 1:34:02 PM
Acinetobacter_baumannii_snp_di...	1 KB	8/12/2026 1:34:03 PM
Acinetobacter_baumannii_snps_a...	389 KB	8/12/2026 1:34:03 PM

File Explorer view showing the contents of the directory `.../analysis/corge0928/output-20260812164858/Acinetobacter_baumannii/`.

Name	Size	Changed
..		8/12/2026 1:31:52 PM
assemblies		8/12/2026 1:31:52 PM
genomic_context_groups		8/12/2026 1:48:53 PM
linkages		8/12/2026 1:34:04 PM
mashtree		8/12/2026 1:37:56 PM
microreact		8/12/2026 1:48:51 PM
parsnp		8/12/2026 1:34:03 PM
poodle_samplesheets		8/12/2026 1:48:55 PM
ReporTree		8/12/2026 1:48:51 PM

SNP distance matrix output from `snp-dists 1.2.0`.

snp-dists 1.2.0	SRR40064	SRR40064	SRR40064	SRR40064	SRR40064142
SRR40064153	0	51486	51606	54514	51468
SRR40064547	51486	0	462	51542	57
SRR40064574	51606	462	0	51512	443
SRR40064187	54514	51542	51512	0	51522
SRR40064142	51468	57	443	51522	0

APPLICATION CONT.

./../output-20260812164858/Acinetobacter_baumannii/poodle_samplesheets/

Name	Size	Changed
..		8/12/2026 1:48:55 PM
Acinetobacter_baumannii_poodl...	1 KB	8/12/2026 1:48:55 PM

.../analysis/corge0928/output-20260812164858/Acinetobacter_baumannii/

Name	Size	Changed
..		8/12/2026 1:31:52 PM
assemblies		8/12/2026 1:31:52 PM
genomic_context_groups		8/12/2026 1:48:53 PM
linkages		8/12/2026 1:34:04 PM
mashtree		8/12/2026 1:37:56 PM
microreact		8/12/2026 1:48:51 PM
parsnp		8/12/2026 1:34:03 PM
poodle_samplesheets		8/12/2026 1:48:55 PM
ReporTree		8/12/2026 1:48:51 PM

sample,fastq_1,fastq_2,annotation,assembly,cluster_id,species,reference
SRR40064142,,,/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/corge0928/output/Acinetobacter_baumannii/assemblies/SRR40064142.fna,HC57-C1,Acinetobacter_baumannii,/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/corge0928/output/Acinetobacter_baumannii/assemblies/SRR40064142.fna
SRR40064547,,,/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/corge0928/output/Acinetobacter_baumannii/assemblies/SRR40064547.fna,HC57-C1,Acinetobacter_baumannii,/blue/bphl-florida/n.yengalareddy/repos/bphl-molecular/analysis/corge0928/output/Acinetobacter_baumannii/assemblies/SRR40064142.fna

APPLICATION CONT.

.../corge0928/output-20260812164858/Acinetobacter_baumannii/ReportTree/

Name	Size	Changed
..		8/12/2026 1:48:55 PM
Acinetobacter baumannii.log	2 KB	8/12/2026 1:48:51 PM
Acinetobacter_baumannii_cluster...	301 KB	8/12/2026 1:48:51 PM
Acinetobacter_baumannii_partiti...	137 KB	8/12/2026 1:48:51 PM
Acinetobacter_baumannii_single...	1 KB	8/12/2026 1:48:51 PM

.../analysis/corge0928/output-20260812164858/Acinetobacter_baumannii/

Name	Size	Changed
..		8/12/2026 1:31:52 PM
assemblies		8/12/2026 1:31:52 PM
genomic_context_groups		8/12/2026 1:48:53 PM
linkages		8/12/2026 1:34:04 PM
mashtree		8/12/2026 1:37:56 PM
microreact		8/12/2026 1:48:51 PM
parsnp		8/12/2026 1:34:03 PM
poodle_samplesheets		8/12/2026 1:48:55 PM
ReportTree		8/12/2026 1:48:51 PM

#partition	cluster	cluster_length	samples
single-0x1.0	singleton_1	1	SRR40064153
single-0x1.0	singleton_2	1	SRR40064547
single-0x1.0	singleton_3	1	SRR40064574
single-0x1.0	singleton_4	1	SRR40064187
single-0x1.0	singleton_5	1	SRR40064142
single-1x1.0	singleton_1	1	SRR40064153
single-1x1.0	singleton_2	1	SRR40064547
single-1x1.0	singleton_3	1	SRR40064574
single-1x1.0	singleton_4	1	SRR40064187
single-1x1.0	singleton_5	1	SRR40064142
single-2x1.0	singleton_1	1	SRR40064153
single-2x1.0	singleton_2	1	SRR40064547
single-2x1.0	singleton_3	1	SRR40064574
single-2x1.0	singleton_4	1	SRR40064187
single-2x1.0	singleton_5	1	SRR40064142
single-3x1.0	singleton_1	1	SRR40064153
single-3x1.0	singleton_2	1	SRR40064547
single-3x1.0	singleton_3	1	SRR40064574
single-3x1.0	singleton_4	1	SRR40064187
single-3x1.0	singleton_5	1	SRR40064142
single-4x1.0	singleton_1	1	SRR40064153
single-4x1.0	singleton_2	1	SRR40064547
single-4x1.0	singleton_3	1	SRR40064574
single-4x1.0	singleton_4	1	SRR40064187
single-4x1.0	singleton_5	1	SRR40064142
single-5x1.0	singleton_1	1	SRR40064153

sequence	single-0x1.0	single-1x1.0	single-2x1.0	single-3x1.0	single-4x1.0	single-5x1.0	single-6x1.0	single-7x1.0	single-8x1.0	single-9x1.0	single-10x1.0	single-11x1.0	single-12x1.0	single-13x1.0	single-14x1.0
SRR40064153	singleton_1	singleton_1	singleton_1	singleton_1	singleton_1	singleton_1	singleton_1	singleton_1	singleton_1	singleton_1	singleton_1	singleton_1	singleton_1	singleton_1	singleton_1
SRR40064547	singleton_2	singleton_2	singleton_2	singleton_2	singleton_2	singleton_2	singleton_2	singleton_2	singleton_2	singleton_2	singleton_2	singleton_2	singleton_2	singleton_2	singleton_2
SRR40064574	singleton_3	singleton_3	singleton_3	singleton_3	singleton_3	singleton_3	singleton_3	singleton_3	singleton_3	singleton_3	singleton_3	singleton_3	singleton_3	singleton_3	singleton_3
SRR40064187	singleton_4	singleton_4	singleton_4	singleton_4	singleton_4	singleton_4	singleton_4	singleton_4	singleton_4	singleton_4	singleton_4	singleton_4	singleton_4	singleton_4	singleton_4
SRR40064142	singleton_5	singleton_5	singleton_5	singleton_5	singleton_5	singleton_5	singleton_5	singleton_5	singleton_5	singleton_5	singleton_5	singleton_5	singleton_5	singleton_5	singleton_5

C O N C L U S I O N

-  Fundamentals of CorGe+
-  Installation and setup of CorGe+ in HPG
-  Successfully executed job query for CorGe+
-  Generated output files



Advanced Molecular Detection

Southeast Region Bioinformatics

Questions?

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